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Systems and methods for manufacturing light fixtures

Abstract

Systems and methods for manufacturing light fixtures. A data processing system can identify a plurality of attributes of a lighting system. The plurality of attributes can comprise at least one of a luminescent type, a mounting type, or a luminescent feature. The data processing system can detect, responsive to input via a graphical user interface, a layout for installation of the lighting system. The data processing system can select, using a model configured to satisfy a gap constraint, a plurality of light fixtures based on the layout and the plurality of attributes that, when electrically coupled to conform to the layout, satisfy the gap constraint. The data processing system can provide an instruction to cause manufacturing of the selected plurality of light fixtures in accordance with the layout and the plurality of attributes.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) This disclosure relates generally to systems and methods for manufacturing light fixtures. In particular, the systems and methods of this disclosure can generate a light fixture configuration according to a layout for a lighting system.

BACKGROUND

(2) Users can select desired features of light fixtures from a specification provided by an entity (e.g., manufacturer or distributor). The users can provide the selected features in the specification to the manufacturer for manufacturing the light fixtures. Based on the selected features, the manufacturer may manufacture the light fixtures for the installation of a lighting system.

BRIEF SUMMARY OF THE DISCLOSURE

(3) Systems and methods of this disclosure are directed to manufacturing light fixtures. Entities, such as manufacturers or distributors, can provide end users with a predefined or static specification including selectable attributes for a lighting system. The attributes may include types of products, types of technology (e.g., luminescent type), type of mount, power consumption (e.g., per foot of lighting fixture), dimension of lighting fixture, color rendering index (CRI), or color temperature (CCT), among others. The users can select desired attributes of the light fixture to request for production or distribution from the entity. However, it may be challenging to accurately determine the compatibility between the different attributes and select these attributes based on the static specification. In particular, the static specification may include all information related to all types of light fixtures, even information unrelated to certain attributes selected by the user. For example, the user may select a certain mounting type for the light fixtures and various features of the light fixtures for installation using the mounting type. Depending on the mounting type, certain features may be available or unavailable. In this case, the user may wait for approval from the entity given the potential

compatibility issues, thereby consuming extensive time and excessive resources (e.g., network resources from iterative communications, storage/memory for storing specifications with unnecessary information, etc.) to finalize the configuration of the light fixtures and initiate production or distribution. Further, although light fixtures with the specified features may be available, the user may select certain lengths and types of light fixtures that cause discrepancies with the desired layout for a lighting system installation, such as undesirable gaps between the light fixtures, insufficient power feeds, etc.

(4) The systems and methods of this technical solution can include a data processing system configured to dynamically provide compatible attributes or features depending on the selection of one or more attribute(s) and select various light fixtures according to a configurable layout for lighting system installation and the selected attributes to satisfy a gap constraint. For example, the data processing system can provide a graphical user interface (GUI) to a client device of the user via an application launched on the client device. The data processing system can receive at least one of the type of product (e.g., compose, graphite, runway, etc.) or the type of technology (e.g., tunable, static, dim-warm, sunset-dim, etc.) for the lighting system selected via the GUI. The data processing system can determine the types of attributes (e.g., power, voltage, CCT, CRI, finish, control, etc.) that are available according to the selected type of product or type of technology. The data processing system can adjust the available types of attributes in response to certain attribute(s) being selected or changes made to any of the selections for compatibility between the attributes. The data processing system can store the selected attributes as part of a configuration for light fixtures in a data repository.

(5) In further example, the data processing system can detect a layout for installation of the lighting system input by the user via the GUI. The data processing system can select the various light fixtures according to the layout and the configuration (or other configuration(s) specified by the user) that satisfies a gap constraint when the light fixtures are electrically coupled. The gap constraint can refer to the maximum distance between the light fixtures. The data processing system can select different lengths or shapes of light fixtures having the attributes to satisfy the layout and the gap constraint. Hence, by utilizing the data processing system to dynamically select the light fixtures that conform with the specified layout, gap constraint, and compatible attributes, the systems and methods can digitize and expedite the specification process, thereby avoiding erroneous selections of incompatible features, enhancing user experience in the selection and installation of the lighting system via the configurable layout, and reducing resource consumption from iterative communications between the user and the entity when selecting features using the static specification.

(6) In one aspect, this disclosure is directed to a method for manufacturing light fixtures. The method can include identifying, by a data processing system comprising one or more processors and memory, a plurality of attributes of a lighting system. The plurality of attributes can include at least one of a luminescent type, a mounting type, or a luminescent feature. The method can include detecting, by the data processing system responsive to input via a graphical user interface, a layout for installation of the lighting system. The method can include selecting, by the data processing system using a model configured to satisfy a gap constraint, a plurality of light fixtures based on the layout and the plurality of attributes that, when electrically coupled to conform to the layout, satisfy the gap constraint. The method can include providing, by the data processing system, an instruction to cause manufacturing of the selected plurality of light fixtures in accordance with the layout and the plurality of attributes.

(7) The method can include providing, by the data processing system via the graphical user interface, a digital blueprint of a physical space in which the lighting system is to be installed. The method can include detecting, by the data processing system responsive to input via the graphical user interface, a plurality of coordinates on the digital blueprint for installation of the lighting system. The method can include generating, by the data processing system, the layout based on the plurality of coordinates on the digital blueprint.

(8) The method can include detecting, by the data processing system responsive to input via the graphical user interface, the layout comprising a first dimension and a second dimension that is perpendicular with the first dimension. The method can include selecting, by the data processing system, the plurality of light fixtures comprising a first light fixture with a first run size and a straight run, a second light fixture with a second run size and an L-shape, and a third light fixture with a third size and the straight run. The first light fixture can be electrically coupled to the third light fixture via the second light fixture.

(9) The method can include selecting, by the data processing system, the model from a plurality of models based on at least one of the plurality of attributes. The method can include selecting, by the data processing system, the model from a plurality of models based on the mounting type and the luminescent feature.

(10) The method can include selecting, by the data processing system, the model to satisfy a unit of energy per distance ratio. The method can include selecting, by the data processing system, the plurality of light fixtures comprising a first light fixture and a second light fixture having a different configuration than the first light

fixture. The different configuration can include at least one of a length or a shape.

(11) The method can include selecting, by the data processing system, at least a portion of the plurality of light fixtures responsive to detecting, via the graphical user interface, at least a portion of the layout for installation prior to completion of the layout for installation. The method can include providing, by the data processing system, for render via the graphical user interface prior to completion of the layout for installation, the at least the portion of the plurality of light fixtures.

(12) The method can include identifying, by the data processing system, the plurality of attributes comprising an emergency lighting type. The method can include identifying, by the data processing system, a direction for lighting and a number of power feeds for the layout. The method can include selecting, by the data processing system, the plurality of light fixtures based on the direction and the number of power feeds.

(13) The method can include determining, by the data processing system, the plurality of light fixtures selected via the model exceeds a power threshold. The method can include adding, by the data processing system, a second power feed responsive to the determination.

(14) In another aspect, this disclosure is directed to a system for manufacturing light fixtures. The system can include a data processing system comprising one or more processors, coupled with memory. The data processing system can identify a plurality of attributes of a lighting system. The plurality of attributes can include at least one of a luminescent type, a mounting type, or a luminescent feature. The data processing system can detect, responsive to input via a graphical user interface, a layout for installation of the lighting system. The data processing system can select, via a model configured to satisfy a gap constraint, a plurality of light fixtures based on the layout and the plurality of attributes that, when electrically coupled to conform to the layout, satisfy the gap constraint. The data processing system can provide an instruction to cause manufacturing of the selected plurality of light fixtures in accordance with the layout and the plurality of attributes.

(15) The data processing system can provide, via the graphical user interface, a digital blueprint of a physical space in which the lighting system is to be installed. The data processing system can detect, responsive to input via the graphical user interface, a plurality of coordinates on the digital blueprint for installation of the lighting system. The data processing system can generate the layout based on the plurality of coordinates on the digital blueprint. For example, the data processing system can perform automated drawing creation from existing PDF or CAD files. To do so, the data processing system can receive, from a user upload, a drawing and a selection of a line on the drawing that is intended by the architect to depict a lighting type. The data processing system can then identify and converting that bitmap or vector into dimensions and subsequent light fixtures.

(16) The data processing system can detect, responsive to input via the graphical user interface, the layout comprising a first dimension and a second dimension that is perpendicular with the first dimension. The data processing system can select the plurality of light fixtures comprising a first light fixture with a first run size and a straight run, a second light fixture with a second run size and an L-shape, and a third light fixture with a third size and the straight run. The first light fixture can be electrically coupled to the third light fixture via the second light fixture.

(17) The data processing system can select the model from a plurality of models based on at least one of the plurality of attributes. The data processing system can select the plurality of light fixtures comprising a first light fixture and a second light fixture having a different configuration than the first light fixture.

(18) The data processing system can determine the plurality of light fixtures selected via the model exceeds a power threshold. The data processing system can add a second power feed responsive to the determination.

(19) In yet another aspect, this disclosure is directed to a non-transitory computer readable storage medium. The non-transitory computer readable storage medium can store instructions that, when executed by one or more processors, cause the one or more processors to identify a plurality of attributes of a lighting system. The plurality of attributes can include at least one of a luminescent type, a mounting type, or a luminescent feature. The instructions can cause the one or more processors to detect, responsive to input via a graphical user interface, a layout for installation of the lighting system. The instructions can cause the one or more processors to select, via a model configured to satisfy a gap constraint, a plurality of light fixtures based on the layout and the plurality of attributes that, when electrically coupled to conform to the layout, satisfy the gap constraint. The instructions can cause the one or more processors to provide an instruction to cause manufacturing of the selected plurality of light fixtures in accordance with the layout and the plurality of attributes.

(20) The instructions can include instructions to provide, via the graphical user interface, a digital blueprint of a physical space in which the lighting system is to be installed. The instructions can include instructions to detect, responsive to input via the graphical user interface, a plurality of coordinates on the digital blueprint for installation of the lighting system. The instructions can include instructions to generate the layout based on the plurality of coordinates on the digital blueprint.

(21) These and other aspects and implementations are discussed in detail below. The foregoing information and the following detailed description include illustrative examples of various aspects and implementations, and provide an overview or framework for understanding the nature and character of the claimed aspects and implementations. The drawings provide illustration and a further understanding of the various aspects and implementations, and are incorporated in and constitute a part of this specification.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) The accompanying drawings are not intended to be drawn to scale. Like reference numbers and designations in the various drawings indicate like elements having similar structure or functionality. For purposes of clarity, not every component may be labeled in every drawing. In the drawings:
- (2) FIG. 1 is a block diagram of an example system for manufacturing light fixtures, in accordance with an implementation;
- (3) FIGS. 2-9 are example GUIs to select the light fixtures for a lighting system, in accordance with an implementation;
- (4) FIG. 10 is an example flow diagram of a method for manufacturing light fixtures, in accordance with an implementation; and
- (5) FIG. 11 is a block diagram illustrating an architecture for a computer system that can be employed to implement elements of the systems and methods described and illustrated herein, including, for example, aspects of the system depicted in FIG. 1, and the operations depicted in FIGS. 2-10.
- (6) The features and advantages of the present solution will become more apparent from the detailed description set forth below when taken in conjunction with the drawings, in which like reference characters identify corresponding elements throughout. Other features, aspects, and advantages of the subject matter will become apparent from the description, the drawings, and the claims.

DETAILED DESCRIPTION

- (7) Following below are more detailed descriptions of various concepts related to, and implementations of, methods, apparatuses, and systems of manufacturing light fixtures. The various concepts introduced above and discussed in greater detail below may be implemented in any of numerous ways.
- (8) Lighting manufacturers or distributors (e.g., entities) can provide light fixtures to users according to the attributes of a lighting system selected in a specification (e.g., light fixture specification predefined by the manufacturer or distributor). The specification provided by these entities may be static, including all attributes or other information regarding various types of lighting fixtures. However, certain attributes may not be compatible with each other or may not be available because of, for instance, power restrictions between different types of technologies, available attributes associated with certain types of products or technologies, available lengths for certain types of technology, etc. Further, even with the user-selected attributes for the light fixtures, electrical connections between these light fixtures may not conform to the desired layout for lighting system installation due to unaccounted gaps between the light fixtures, power limit for each run of light fixtures, or inaccurate length estimation by the user, to list a few. To address the error-prone process of selecting attributes/configurations for the lighting system from the static specification, the system can digitize the specification by providing a graphical user interface (GUI) to a client device configured to receive user input and dynamically display available or compatible attributes responsive to the selected attribute(s). The system can provide a digital blueprint via the GUI to enable the user to generate, customize, or otherwise update the desired layout of the lighting system. The system can use a model configured to satisfy various constraints to generate the layout, such as gap constraint, light fixture availability constraint, configuration constraint, or power constraint, among others. Thus, the system can expedite the specification process by digitizing the specification using the model, thereby avoiding the erroneous selection of attributes, enhancing user experience in the selection and installation of the lighting system via the configurable layout, and reducing resource consumption from iterative communications between the user and the entity to obtain approval for the light fixtures.
- (9) Referring now to FIG. 1, a block diagram of an example system **100** for manufacturing light fixtures is shown. The system **100** can include at least one network **102**, at least one data processing system **104**, at least one client device **106**, and at least one manufacturing device **108**. The components (e.g., network **102**, data processing system **104**, client device **106**, or manufacturing device **108**) of the system **100** can include or be composed of hardware, software, or a combination of hardware and software components. The one or more

components (e.g., data processing system **104**, client device **106**, or manufacturing device **108**) of the system **100** can establish communication channels or transfer data via the network **102**. For example, the client device **106** can communicate with at least one of the data processing system **104** or the manufacturing device **108** via the network **102**. In another example, the data processing system **104** can communicate with other devices, such as the client device **106** or the manufacturing device **108** via the network **102**. The communication channel between various different network devices can communicate with each other via the network **102** or different networks **102**.

(10) The network **102** can include computer networks such as the Internet, local, wide, metro or other area networks, intranets, satellite networks, other computer networks such as voice or data mobile phone communication networks, and combinations thereof. The network **102** may be any form of computer network that can relay information between the one or more components of the system **100**. The network **102** can relay information between client devices **106** and one or more information sources, such as web servers or external databases, amongst others. In some implementations, the network **102** may include the Internet and/or other types of data networks, such as a local area network (LAN), a wide area network (WAN), a cellular network, a satellite network, or other types of data networks. The network **102** may also include any number of computing devices (e.g., computers, servers, routers, network switches, etc.) that are configured to receive and/or transmit data within the network **102**. The network **102** may further include any number of hardwired and/or wireless connections. Any or all of the computing devices described herein (e.g., data processing system **104**, client device **106**, or manufacturing device **108**) may communicate wirelessly (e.g., via Wi-Fi, cellular, radio, etc.) with a transceiver that is hardwired (e.g., via a fiber optic cable, a CAT5 cable, etc.) to other computing devices in the network **102**. Any or all of the computing devices described herein (e.g., data processing system **104**, client device **106**, or manufacturing device **108**) may also communicate wirelessly with the computing devices of the network **102** via a proxy device (e.g., a router, network switch, or gateway).

(11) The client device **106** can include at least one processor and a memory, e.g., a processing circuit. The client device **106** can include various hardware or software components, or a combination of both hardware and software components. The client device **106** can be constructed with hardware or software components. For example, the client device **106** can include, but is not limited to, a television device, a mobile device, smart phone, personal computer, a laptop, a gaming device, a kiosk, or any other type of computing device. The client device **106** can be operated, used, or accessible by the user.

(12) The client device **106** can include at least one interface for establishing a connection to the network **102**. The client device **106** can communicate with other components of the system **100** via the network **102**, such as the manufacturing device **108** or the data processing system **104**. For example, the client device **106** can communicate data packets with the manufacturing device **108** via the network **102**. The client device **106** can communicate with the data processing system **104** via the network **102**. In some cases, the client device **106** can communicate with other client devices **106** communicatively coupled to the network **102**.

(13) The client device **106** can include, store, execute, or maintain various application programming interfaces (“APIs”) in the memory (e.g., local to the client device **106**). The APIs can include or be any types of API, such as Web APIs (e.g., open APIs, Partner APIs, Internal APIs, or composite APIs), web server APIs (e.g., Simple Object Access Protocol (“SOAP”), XML-RPC (“Remote Procedure Call”), JSON-RPC, Representational State Transfer (“REST”)), among other types of APIs or protocol. The client device **106** can use at least one of various protocols for communication with the data processing system **104**. The protocol can include at least a transmission control protocol (“TCP”), a user datagram protocol (“UDP”), or an internet control message protocol (“ICMP”). The communicated data can include a message, a content, a request, or otherwise information to be transmitted from the client device **106** to other devices in the network **102**.

(14) The manufacturing device **108** can be a computing device operated by an entity (e.g., manufacturer or distributor). The manufacturing device **108** can be in electrical communication with equipment for manufacturing light fixtures. In some cases, the manufacturing device **108** can be in electrical communication with equipment for obtaining, gathering, sorting, packing, or otherwise distributing the light fixtures to the user. The manufacturing device **108** can be composed of hardware or software components, or a combination of both hardware or software components. The manufacturing device **108** can be in communication with the data processing system **104** via the network **102**. The manufacturing device **108** can be in communication with the client device **106** via the network **102**.

(15) The manufacturing device **108** can receive various instructions from the data processing system **104** or the client device **106**. The instruction can include manufacturing one or more light fixtures selected or requested by the user. For example, in response to receiving the instruction, the manufacturing device **108** can command at least one manufacturing equipment to initiate a manufacturing process for the one or more light fixtures. The

instruction can include sorting or grouping the light fixtures according to a particular run of light fixtures, the lighting system of a project, types of selected light fixtures, or other organization categories specified by the user. For example, in response to receiving the instruction, the manufacturing device **108** can command the distribution equipment to collect the selected light fixtures (e.g., stored in physical storage space or inventory of the entity) and pack the light fixtures for distribution to the user.

(16) The manufacturing device **108** can communicate with at least the data processing system **104** or the client device **106** to update the availability of light fixtures. The manufacturing device **108** can communicate with at least the data processing system **104** or the client device **106** to update compatibilities between attributes of the lighting system (e.g., including luminescent features of the light fixtures). In some cases, the manufacturing device **108** can push or send the updates to an application used by the data processing system **104** or the client device **106** to access the GUI for selecting attributes or generating the layout of the lighting system. For instance, the manufacturing device **108** can be in communication with the data processing system **104** and the client device **106** via the application. The application can be accessed or modified by an administrator (e.g., entity) of the manufacturing device **108**. The manufacturing device **108** can update the light fixture availability and attributes compatibilities in real-time via the application, such as in response to light fixtures added or removed from the inventory of the entity. The updates can be reflected in the application accessed by the client device **106** and shown to the user via the GUI of the application.

(17) In some cases, the manufacturing device **108** can delegate management of the application (e.g., updating product availability or attribute compatibility, etc.) to the data processing system **104**. For instance, subsequent to receiving instructions to manufacture or distribute a number of light fixtures from the data processing system **104**, the manufacturing device **108** can respond or send an acknowledgment to the data processing system **104**, thereby allowing the data processing system **104** to remove the specified number of light fixtures from availability. In another example, the data processing system **104** can receive input data from the administrator indicating adjusted compatibilities between different attributes, such as adding compatibility support or removing compatibility support. In this case, the data processing system **104** can transmit the indication of adjusted compatibilities to the data processing system **104**, enabling the data processing system **104** to update compatibilities between the attributes in the application. In some arrangements, the manufacturing device **108** can include or be a part of the data processing system **104** to perform certain features similar to the data processing system **104**.

(18) The data processing system **104** can include various components to digitize the specification for the user and provide a digital blueprint for configuring and visualizing the layout of the lighting system in a physical space. The data processing system **104** can include at least one interface **110**, at least one user interface generator **112**, at least one layout configurator **114**, at least one light fixture selector **116**, at least one model manager **118**, and at least one data repository **120**. The data repository **120** can include at least one user interface storage **122**, at least one attribute storage **124**, at least one layout storage **126**, at least one light fixture storage **128**, at least one constraint storage **130**, and at least one model storage **132**. Individual components (e.g., interface **110**, user interface generator **112**, layout configurator **114**, light fixture selector **116**, model manager **118**, or data repository **120**) of the data processing system **104** can be composed of hardware, software, or a combination of hardware and software components. Individual components of the data processing system **104** can be in electrical communication with each other. For instance, the interface **110** can exchange data or communicate with the user interface generator **112**, layout configurator **114**, light fixture selector **116**, model manager **118**, or data repository **120**. The one or more components of the data processing system **104** can be used to perform features or functionalities, such as generating the GUI, configuring the layout for the lighting system, selecting the light fixtures for the lighting system, managing the models, or storing lighting systems configured by the users via the GUI. The data processing system **104** can operate remotely from the client device **106**, the manufacturing device **108**, or other devices in the system **100**. In some cases, the data processing system **104** can operate locally to the manufacturing device **108** or certain other devices in the system **100**.

(19) In some cases, the data processing system **104** can be a part of the manufacturing device **108**, such as an integrated device, embedded device, a server-operated device, or a device accessible by the administrator of the manufacturing device **108**. For example, the data processing system **104** can perform operations local or on-premise to the manufacturing device **108**, such as processing inputs from the client device **106**. One or more components (e.g., interface **110**, user interface generator **112**, layout configurator **114**, light fixture selector **116**, or model manager **118**) of the data processing system **104** can be executed on the manufacturing device **108**. In some cases, one or more components or functions of the data processing system **104** can be packaged into a script, agent, or bot configured to execute on the manufacturing device **108** or other devices connected to the

network **102**.

(20) The interface **110** can interface with the network **102**, devices within the system **100** (e.g., client devices **106** or manufacturing devices **108**), or components of the data processing system **104**. The interface **110** can include features and functionalities to interface with the aforementioned components. For example, the interface **110** can include standard telephone lines LAN or WAN links (e.g., 802.11, T1, T3, Gigabit Ethernet, Infiniband), broadband connections (e.g., ISDN, Frame Relay, ATM, Gigabit Ethernet, Ethernet-over-SONET, ADSL, VDSL, BPON, GPON, fiber optical including FiOS), wireless connections, or some combination of any or all of the above. Connections can be established using a variety of communication protocols (e.g., TCP/IP, Ethernet, ARCNET, SONET, SDH, Fiber Distributed Data Interface (FDDI), IEEE 802.11a/b/g/n/ac CDMA, GSM, WiMax and direct asynchronous connections). The interface **110** can include at least a built-in network adapter, network interface card, PCMCIA network card, EXPRESSCARD network card, card bus network adapter, wireless network adapter, USB network adapter, modem, or any other device suitable for interfacing one or more devices within the system **100** to any type of network capable of communication. The interface **110** can communicate with one or more aforementioned components to receive data from the client devices **106**, such as selections of attributes, coordinates of the digital blueprint of physical space, etc. The interface **110** can communicate with one or more aforementioned components to receive data from the manufacturing device **108**, such as updates to the attributes (e.g., compatibility, types, values, etc.), updates on light fixture availability in the inventory, or updates on supporting light fixtures for manufacturing, among others.

(21) The user interface generator **112** can generate, provide, or update GUI for display via at least one of the client device **106** or the manufacturing device **108**. The user interface generator **112** can provide the GUI via the application accessible by the client device **106** or the manufacturing device **108**. The user interface generator **112** can generate the GUI depending on the authority granted to specific users. For example, a GUI for the administrator having administrator privilege can include interactive elements for adjusting at least the types of attributes for light fixtures or the attribute values (e.g., values of luminescent features, such as a unit of energy per distance ratio, dimensions of light fixtures, CCT value, etc.). The GUI for the administrator can include other information accessible by lower-level users. In another example, the GUI for the lower-level users, such as customers, can include interactive elements for selecting attributes for the lighting system. In response to the user selecting at least one interactive element, the user interface generator **112** may update the visibility or availability of certain other attributes (e.g., types of luminescent features) depending on another attribute being selected (e.g., luminescent type or mounting type). The user interface generator **112** can generate other information for displaying the GUI depending on at least the page of the application (e.g., dashboard, project creation, account, specification panel, etc.), such as shown in conjunction with at least one of FIGS. 2-9, for example.

(22) The user interface generator **112** can generate a digital blueprint of a physical space in which the lighting system is to be installed. The digital blueprint refers to a virtual space or digital canvas in the application that allows the user to specify at least one of the dimensions (e.g., length, width, or height) of the lighting system, types of light fixtures to use, locations of the light fixtures, direction of lighting, shapes of the light fixtures, power feed (e.g., power input or power outlet) location, or emergency light position, among other lighting system configuration.

(23) The user interface generator **112** can generate or update the GUI to display the layout configured by the layout configurator **114** responsive to the user input. The user interface generator **112** can generate or update the GUI to display visual representations of light fixtures selected by the light fixture selector **116** according to the selected attributes. The user interface generator **112** can generate or update other elements of the GUI, such as responsive to user inputs from the client device **106** or updates from the manufacturing device **108**, among other authorized devices within the network **102**.

(24) The layout configurator **114** can generate, update, or otherwise configure the layout for installation of the lighting system. The layout configurator **114** can generate the layout based on inputs from the client device **106** via the GUI. The layout configurator **114** can update the layout in response to inputs via the GUI. For example, in response to input via the GUI, the layout configurator **114** can detect various coordinates on the digital blueprint for the installation of the lighting system. The coordinates can include latitudinal and longitudinal coordinates on the digital blueprint. The layout configurator **114** determines the sequence of coordinates input by the user. The layout configurator **114** can generate the layout based on the coordinates on the digital blueprint. The layout can include visual representations of the light fixtures extending between various coordinates according to the sequence of coordinates, such as a first set of light fixture(s) from a first coordinate to a second coordinate, a second set of light fixture(s) from the second coordinate to a third coordinate, and so forth. The layout configurator **114** can provide at least a portion of the layout or a portion of

the coordinates to the light fixture selector **116** to select the light fixtures. The layout configurator **114** can receive the selected light fixtures from the light fixture selector **116**. The layout configurator **114** (or the light fixture selector **116**) can send the layout including the light fixtures to the user interface generator **112** for rendering.

(25) The layout configurator **114** can allow at least one layout within the digital blueprint. In some cases, the layout configurator **114** can allow multiple layouts within a single digital blueprint. Each layout can include the light fixtures consecutively connected or positioned next to each other within a specified gap constraint. With multiple layouts, the layout configurator **114** can receive multiple sequences of coordinates input by the user in the digital blueprint.

(26) The layout configurator **114** can detect a portion or a subset of a layout input via the graphical user interface prior to completing the layout for installation. The layout configurator **114** can send the portion of the layout to the light fixture selector **116** for selecting the light fixtures according to the provided portion in response to the detection. In this case, the selected light fixtures can be a subset of various light fixtures to complete the layout. The layout configurator **114** or the light fixture selector **116** can send or provide at least the portion of the light fixtures to the user interface generator **112** for rendering via the GUI before completing the layout for installation.

(27) The layout configurator **114** can identify various attributes for generating the layout. The attributes can be provided or updated according to input via the GUI for the digital blueprint. For example, the attribute can include an emergency lighting type. The layout configurator **114** can receive an indication that the emergency lighting type is selected. The layout configurator **114** can detect coordinates in response to input via the GUI for the emergency lighting type. Subsequently, the layout configurator **114** can generate a layout for emergency lighting type. The layout configurator **114** can request the selection of the emergency light fixture from the light fixture selector **116**. The layout configurator **114** can provide the layout of the emergency light to the user interface generator **112** for rendering a visual representation of the emergency light fixtures extending across various coordinates.

(28) The layout configurator **114** can receive input via the GUI rendering the digital blueprint indicating the direction for lighting. For example, the lighting direction may default to face either up or down for a horizontal layout of light fixtures. In another example, the lighting direction may default to face left or right for a vertical layout of light fixtures. The layout configurator **114** can receive input via the GUI selecting at least a portion of the layout and an indication to switch the lighting direction for the portion of the layout. The layout configurator **114** can transmit any layout update to the user interface generator **112** for updating the GUI or rendering of the lighting fixtures.

(29) The layout configurator **114** can insert or add a power feed at the first coordinate input by the user. The layout configurator **114** can detect inputs via the GUI to update the power feed location or add at least one additional power feed along a portion of the layout. The layout configurator **114** can update the power feed location or include at least one additional power feed based on the input.

(30) In some cases, the layout configurator **114** can use at least one model managed by the model manager **118** to determine whether various constraints are satisfied in the layout. For example, the layout configurator **114** can use a model to determine whether the light fixtures selected by the light fixture selector **116** satisfy a power constraint. To satisfy the power constraint, the total rated power consumption by the selected light fixtures can be at or below a power threshold. The power threshold can correspond to the maximum power output via the power feed or the power outlet. If the light fixtures exceed or do not satisfy the power constraint, the layout configurator **114** can dynamically add at least one additional power feed (e.g., a second power feed) to a portion of the layout, such that the one or more light fixtures that exceed the power constraint can be supplied with power via the second power feed.

(31) The layout configurator **114** can use the model to configure the layout to satisfy other constraints, such as a gap constraint. The gap constraint can refer to the maximum allowable non-illuminated distance between light fixtures for a given run of light fixtures. The gap constraint can be predefined as 0.5 inches, 0.4 inches, 0.3 inches, etc. The selection of the light fixtures and positioning of the selected light fixtures in the layout can adhere to the gap constraint using the model. Exceeding the gap constraint can result in visually noticeable gaps or separations between illuminated light fixtures. The light fixture selector **116** or other components of the data processing system **104** can use the model to perform their respective features or operations to satisfy the various constraints.

(32) For example, the layout configurator **114** can provide the layout to the light fixture selector **116** for selecting, using the model, the light fixtures with respective types or lengths that satisfy the layout and the gap constraint. The layout configurator **114** can receive the selected light fixtures from the light fixture selector **116**.

The layout configurator **114** can position the selected light fixtures according to the layout and gap constraint. The layout configurator **114** can provide the layout with the light fixtures to the user interface generator **112** for rendering the light fixtures in the layout via the GUI.

(33) The light fixture selector **116** can select the light fixtures according to the attributes selected by the user and the layout. The light fixture selector **116** can detect the layout for installation of the lighting system in response to input via the GUI from the user. The light fixture selector **116** can detect the layout generated by the layout configurator **114** according to coordinates received from the client device **106**. The light fixture selector **116** can use the model to select one or more light fixtures to satisfy the gap constraint. The light fixture selector **116** can select the light fixtures based on the layout and the attributes that, when electrically coupled to conform to the layout, satisfy the gap constraint. The attribute can include at least one of a luminescent type, a mounting type, or a luminescent feature. The luminescent type may include at least one of product type, technology type, or other information related to the technology of the light fixture. The mounting type can include at least one of floor mount, wall mount, ceiling mount, low profile mount, adjustable clip, fixed clip, or other types of mounting platform. The luminescent feature can include at least one of luminaire output, power, voltage, CCT, CRI, beam, finish (e.g., color of light fixture), or lighting control, among other features.

(34) The light fixture selector **116** can select the light fixtures including a first light fixture and a second light fixture having a different configuration than the first light fixture. The different configuration can include at least one of a length or a shape. The available length or shape of light fixtures may be different depending on the selected attributes. For example, certain product types may include angled light fixtures and some other product types may not include angled light fixtures. Further, certain types of product or technologies may include a first set of available (e.g., predefined) lengths and other types of product or technologies may include a second set of available lengths, which may or may not include predefined lengths from the first set. For instance, the length can include at least one of 4 inches, 6 inches, 8 inches, 12 inches, 18 inches, 48 inches, etc.

(35) The light fixture selector **116** can use the model to select one or more lengths for the light fixtures to conform with the criteria including the layout and gap constraint. For example, the model may perform simulations of different lengths of light fixtures that conform with the criteria. Take a straight run layout (e.g., lighting system extending from a first coordinate to a second coordinate), for example, the model can select the light fixture(s) with the longest length that is within the layout boundary (e.g., between the first and second coordinates). The model can determine whether these selected light fixture(s) satisfy the criteria by simulating arrangements or positioning of the light fixtures. If at least one arrangement of the selected light fixtures satisfies the criteria, the light fixture selector **116** can use the selected light fixtures for the layout. Otherwise, the light fixture selector **116** can reiterate the process by selecting light fixture(s) having the next longest length that fits the remainder of the layout. After iterating through the available lengths of light fixtures, if the gap constraint is still not satisfied, the light fixture selector **116** can substitute at least one previously selected light fixture with multiple shorter light fixtures, such as replacing the longest light fixture with two relatively shorter light fixtures. For instance, the light fixture selector **116** may replace a 16-inch light fixture with two 8-inch light fixtures, thereby allowing an additional gap between the two 8-inch light fixtures. In further example, the 16-inch light fixture may be replaced with a 4-inch light fixture and two 6-inch light fixtures, allowing two additional gaps between these light fixtures. The total length of the substituted light fixtures may be greater than, less than, or equal to the length of the previous light fixture to satisfy the gap constraint.

(36) The light fixture selector **116** can detect the layout generated by the layout configurator **114** including a first dimension and a second dimension that is perpendicular to, or otherwise angled from, the first dimension. In this case, depending on the selected attribute(s), the light fixture selector **116** can select light fixtures including a first light fixture, a second light fixture, and a third light fixture, for example. The first light fixture can be with a first run size and a straight run. The second light fixture can be with a second run size and an L-shape. The third light fixture can be with a third size and the straight run. The first light fixture can be electrically coupled to the third light fixture via the second light fixture, such that the second light fixture is interposed between the first and third light fixtures. The first light fixture can be perpendicular to the third light fixture. The angle of the second light fixture may be adjustable depending on the selected attributes. For instance, a certain type of product or technology may include light fixtures with 90 degrees angle, while certain other types of product or technology may include light fixtures with an adjustable angle of up to 180 degrees. Different lengths of the angled light fixture may be available depending on the attributes.

(37) The light fixture selector **116** can identify the direction of light according to the layout generated by the layout configurator **114**. The light fixture selector **116** can select the light fixtures to conform with the layout based on the lighting direction. In some cases, the selected light fixture may be bidirectional, such that the user can change the direction of the light fixture during the installation of the lighting system.

(38) The light fixture selector **116** can identify the number of power feed for the layout. The light fixture selector **116** can select the light fixtures based on the number of power feeds. For instance, the light fixture selector **116** can select a number light fixtures configured to connect with the power feed, where the number of these light fixtures corresponds to the number of power feeds. Further, the light fixture selector **116** can select a number of light fixtures with a voided end configured to sit on or cover the power feed (e.g., power input cable connecting to the next light fixture). For example, with two power feeds, the light fixture selector **116** can select at least one light fixture with the voided end. In another example, with three power feeds, the light fixture selector **116** can select at least two light fixtures with the voided end. Having the voided end of the light fixture to cover the exposed power cable can minimize the gap between the light fixtures, satisfying the gap constraint.

(39) In some cases, the light fixture selector **116** can detect that the attribute including an emergency lighting type is selected. Based on the coordinates input via the GUI, the light fixture selector **116** can determine the distance for the emergency light run. The light fixture selector **116** can select one or more emergency light fixtures (e.g., lighting units connected to a battery) to conform with the layout.

(40) Thus, in some cases, the data processing system can perform automated drawing creation from existing PDF or CAD files. To do so, the data processing system can receive, from a user upload, a drawing and a selection of a line on the drawing that is intended by the architect to depict a lighting type. The data processing system can then identify and converting that bitmap or vector into dimensions and subsequent light fixtures.

(41) The model manager **118** can manage models configured to manage compatibilities between different attributes, configure the layout, or select light fixtures to satisfy the constraints, among others. The model can be configured to configure the layout or select the light fixture to satisfy at least one of the selected attributes, gap constraint, power constraint, or other constraints specified by the user or the administrator. The model manager **118** can generate, update, or replace one or more models based on the information from the manufacturing device **108**.

(42) For example, the model manager **118** can receive or obtain the predefined specification from the manufacturing device **108** or the data repository **120**. Using the model, the model manager **118** can identify the attributes of the lighting system available from the entity (e.g., manufacturer or distributor) according to the predefined specification. The attributes can include at least one of a luminescent type, a mounting type, or a luminescent feature. Based on the predefined specification, the model manager **118** can use the model to identify mounting types compatible with each luminescent type. Further, the model can identify the luminescent features compatible with each mounting type or each luminescent type. The model manager **118** can iterate the process for each attribute in the predefined specification to determine compatibilities between all attributes. The model manager **118** may receive an updated predefined specification. Accordingly, the model manager **118** can update compatibilities between the attributes as necessary using the model.

(43) The model manager **118** can select a model from the list of models based on at least one of the attributes selected by the user. For instance, the model manager **118** can select a first model configured to identify attributes compatible or supported with a first selected luminescent type or a second model configured to identify attributes compatible with a second selected luminescent type, etc. Based on the compatibility of the luminescent type, the model can filter attributes that are not compatible or indicate attributes that are compatible.

(44) The model manager **118** can select a model based on at least one of the mounting type or the luminescent feature. For example, the model manager **118** can select a first model configured to identify luminescent features or luminescent types compatible with a first selected mounting type, a second model configured to identify luminescent features or luminescent types compatible with a second selected mounting type, and so forth. In another example, the model manager **118** can select a first model configured to identify mounting types or luminescent types compatible with a first selected set of luminescent feature(s), a second model configured to identify mounting types or luminescent types compatible with a second selected set of luminescent feature(s), and so forth. In some cases, the selected model can be configured to identify compatible or non-compatible attributes associated with a combination of the select mounting type and luminescent feature, among other combinations of attributes.

(45) In some other cases, the model manager **118** can select the model configured to satisfy a unit of energy per distance ratio (e.g., power constraint). The unit of energy per distance ratio can refer to the wattage per foot of light fixtures having the same configuration. In this case, the model is configured to detect or identify the attributes selected by the user to determine a set of light fixtures for selection to satisfy the selected attributes (e.g., configuration). The selected attributes of the light fixtures can reflect the unit of energy per distance ratio for the light fixtures, such as selected CCT, CRI, or lumen output, among others. The model can determine the

total power consumption for the run of light fixtures based on the distance or size of the run and the unit of energy per distance. The model can determine the total available power output via the power feed indicated by the layout configurator **114**. The model can compare the total power consumption and the available power output to determine whether the light fixtures exceed the power constraint. In response to exceeding the power constraint, the model can trigger the layout configurator **114** to input at least one additional power feed or trigger the light fixture selector **116** to select at least one light fixture configured to connect to the power feed or a type of light fixture positioned at the end of the run (e.g., with voided end).

(46) The model manager **118** can provided the selected model to one or more other components of the data processing system **104**, such as the layout configurator **114** or the light fixture selector **116**. The components of the data processing system **104** can use the model to perform their respective functions. For example, the user interface generator **112** can use the model to present, highlight, or filter one or more attributes, such as luminescent types, mounting types, or luminescent features depending on compatibility, availability, or support. The layout configurator **114** can use the model to generate at least a portion of the layout satisfying the various constraints, such as gap constraint, power constraint, or other constraints defined by the administrator or the user. Further, the light fixture selector **116** can use the model to select one or more light fixtures according to the selected attributes (e.g., luminescent type, luminescent feature, mounting type, or emergency lighting type), available lengths, gap constraint, power constraint, conformity to the layout, etc.

(47) In some cases, the model can be used by the light fixture selector **116** for optimizing cost. For instance, one of the selected light fixtures conforming to the layout and gap constraint may include a 16-inch light fixture. If the total cost of two 8-inch light fixtures is cheaper than the 16-inch light fixture, the model can trigger the light fixture selector **116** to replace the 16-inch light fixture with two 8-inch light fixtures instead, depending on lengths available. In certain cases, each of the models can include or correspond to a respective set of scripts or codes configured to perform the features or functionalities described herein.

(48) After finalizing the layout and selecting the light fixtures to conform with the layout, the data processing system **104** can provide or transmit an instruction to the manufacturing device **108** to cause the manufacturing of the selected light fixtures in accordance with the layout and the selected attributes. Causing the manufacturing of the selected light fixtures can refer to notifying, via the manufacturing device **108**, the entity (e.g., manufacturer or distributor) of a new request for light fixtures. According to the request, the entity can activate manufacturing equipment for manufacturing the light fixtures. In some cases, according to the request, the entity can distribute (e.g., pack and ship) the light fixtures to the requester based on the availability of light fixtures in the inventory. In some cases, the manufacturing device **108** may correspond to the manufacturing equipment configured to receive instructions from the data processing system **104**. Accordingly, in response to receiving the instructions, the manufacturing device **108** can initiate a manufacturing process for the requested light fixtures.

(49) The data repository **120** can include the user interface storage **122**. The user interface storage **122** can include, store, or maintain preconfigured user interface (e.g., GUI) for rendering to the client device **106** or the manufacturing device **108** via an application accessible to authorized users. The GUI can include various pages, such as a dashboard, project creation page, project panel including various created project, specification panel including various attributes for selection in a particular project, drawing panel for presenting the digital blueprint or layout, order panel for presenting historical projects ordered by the user, among other pages configurable by the administrator. The GUI can include interactive elements in individual pages, panels, widgets, or tabs in the application. The user can interact with the interactive elements, for example, to navigate between pages, create a project, select attributes for the lighting system, input information, input coordinates in the digital blueprint, or other predefined actions. The GUI can include any other features or information configurable by the administrator, including but not limited to images, texts, pop-up elements, etc. The user interface storage **122** can be accessible by various devices within the network **102** via the application.

(50) The data repository **120** can include the attribute storage **124**. The attribute storage **124** can include, store, or maintain attributes available for the lighting system. The attribute storage **124** can be accessible by the devices within the network **102**, such as the client device **106** or the manufacturing device **108** via the application. The attribute storage **124** can store attributes received from the manufacturing device **108**. The attributes can be updated by the administrator.

(51) The data repository **120** can include the layout storage **126**. The layout storage **126** can include, store, or maintain layouts generated by the client device **106**. For example, in response to finalizing the layout, the user can interact with the GUI to save the configured layout. The layout storage **126** can maintain the layouts generated by the user until the user request to delete certain layouts. In some cases, the layout storage **126** can maintain the layouts until the time associated with each layout expires. The layout storage **126** can be accessed

by the client device **106** or the manufacturing device **108** to add, update, or remove the layout(s).

(52) The data repository **120** can include the light fixture storage **128**. The light fixture storage **128** can include, store, or maintain various types of light fixtures available by the entity. Available light fixtures can refer to light fixtures currently stored in a physical inventory of the manufacturer or distributor. In some cases, available light fixtures can refer to light fixtures supported by the manufacturer for manufacturing purposes. The light fixture storage **128** can be accessed by the client device **106** for light fixture selection. The light fixture storage **128** can be accessed by the manufacturing device **108** to update the available light fixtures for selection by the light fixture selector **116**.

(53) The data repository **120** can include the constraint storage **130**. The constraint storage **130** can include, store, or maintain constraints defined by the administrator or the user. The constraints can include at least one of gap constraint, power constraint, etc. In some cases, the layout configured by the user or selected attributes may be considered part of the constraint. The constraint storage **130** can be accessed by the client device **106** or the manufacturing device **108** via the application for updating, adding, or removing certain constraint(s).

(54) The data repository **120** can include the model storage **132**. The model storage **132** can include, store, or maintain the models selectable by the model manager **118**. The model may be generated by the model manager **118** or the manufacturing device **108**. The model can be updated in response to receiving updated information from the manufacturing device **108**. The model storage **132** can be accessed by the manufacturing device **108**, or other devices within the network **102** for adding, updating, or removing one or more models.

(55) The various storages in the data repository **120** can be accessed by other components of the data processing system **104**. For example, the user interface generator **112** can access the user interface storage **122** to retrieve or obtain GUI for rendering. The layout configurator **114** can access the layout storage **126** to store, modify, remove, or otherwise manage the layouts in the layout storage **126**. The light fixture selector **116** can access the light fixture storage **128** to select at least one of the light fixtures from the light fixture storage **128** satisfying various criteria, such as attributes, gap constraints, etc. The model manager **118** can access the model storage **132** to obtain or select the model. Other examples can be applied, such that the components of the data processing system **104** can access any of the storages in the data repository **120**.

(56) Referring generally to FIGS. 2-9, example GUIs **200-900** to select the light fixtures for a lighting system are shown. The GUIs **200-900** of FIGS. 2-9 can be rendered by the user interface generator **112** on the application executable for the client device **106**, the manufacturing device **108**, or other devices within the network **102**. In some cases, the application can correspond to software or program executed by the data processing system **104**, the client device **106**, or the manufacturing device **108**. For instance, the client device **106** can execute the application to communicate with the data processing system **104** (e.g., user interface generator **112**) to receive data for rendering at least one of the GUIs **200-900**. In this case, the GUIs **200-900** or elements of the GUIs **200-900** can be pre-configured (e.g., by the administrator or developer of the application) and stored in the user interface storage **122**. The elements (e.g., interactive elements, images, texts, objects, click boxes, or other visual representations) of the GUIs **200-900** can be positioned, arranged, or organized by the administrator. The interactive elements can include at least one of buttons, sliders, drop-down menu, etc.

(57) The example GUIs **200-900** of FIGS. 2-9 may include more or fewer details as configured by the administrator. The GUIs **200-900** can represent a certain page or a process for selecting the light fixtures for the lighting system to be requested from the manufacturing device **108**. FIGS. 2-9 can include operations, procedures, or processes executed, performed, or otherwise carried out by one or more components of the system **100** (e.g., data processing system **104**, client device **106**, manufacturing device **108**, or other devices within the network **102**).

(58) Referring now to FIG. 2, in greater detail, the GUI **200** can represent a project creation page for the application. As shown, the GUI **200** for the project creation page can include various input sections for entering information related to the project for installation of the lighting system. The input sections can include user information (e.g., name, address, payment information, etc.), project information (e.g., project name, schedule, etc.), notes for the entity, or other required or optional inputs requested of the user. The GUI **200** can include interactive elements including, but not limited to, at least one of interactive elements **202**, **204**, or **206**. The interactive elements **202**, **204**, or **206** can include or be associated with a hyperlink or a script, where in response to an interaction with the interactive elements **202**, **204**, or **206**, the data processing system **104** (e.g., user interface generator **112**) is configured to redirect the user to a particular page.

(59) For example, the data processing system **104** can redirect, via interaction with the interactive element **202**, the application executing on the client device **106** to a specification page for generating a personalized specification for the lighting system without information irrelevant to the lighting system of the project. Each generated project can include a lighting system configured by the user. The data processing system **104** can

redirect, via interaction with the interactive element **204**, the application to a quote page for generating a quote for the lighting system (e.g., cost for individual light fixtures) associated with the specification. The data processing system **104** can redirect, via interaction with the interactive element **206**, the application to a drawing page for managing layouts for installation of the lighting system associated with the specification or the quote.

(60) By creating a project, the data processing system **104** can provide digitized specification(s) tailored to the specific lighting system configured by the user. The configuration (e.g., selected attributes) of the lighting system may be shared across the specification, quote, and drawings of the project. The specification, quote, and drawings can refer to types of documents within the project. These documents (e.g., specification, quote, or drawing) may share data regarding the configuration of the lighting system with each other. Hence, the updated configuration of the lighting system can be centralized across other documents within the project. The data processing system **104** can store the created project in the data repository **120**. In some cases, the created project may be saved on the respective device executing the application, such as the client device **106** or the manufacturing device **108**, among others.

(61) Referring to FIG. **3**, the GUI **300** can represent a dashboard page (e.g., summary page or overview page) of the application. The GUI **300** can include an overview of a particular document generated by the data processing system **104**. In this case, the GUI **300** can show an example overview of the quote document including one or more projects and details (e.g., notes, costs, type of product, creation time, last modified time, etc.) associated with the projects. The example overview of the quote document can include interactive elements for deleting respective projects. The listing of projects can be interactive elements, where in response to an interaction, the data processing system **104** (e.g., user interface generator **112**) can redirect the application to the corresponding page for editing one of the projects. The GUI **300** can include the interactive elements **202**, **204**, and **206** similarly to GUI **200**. The GUI **300** can include interactive element **302**. In response to an interaction with the interactive element **302**, the data processing system **104** can redirect the application to the project creation page, such as GUI **200** in conjunction with FIG. **2**. Additionally or alternatively, the GUI **300** (or other GUIs **200-900**) can include buttons, tabs, or interactive elements for navigating to different pages within the application.

(62) Referring to FIG. **4**, the GUI **400** can represent a deleted project page. The GUI **400** can include a list of historical projects created by the user (e.g., account owner or authorized personnel) that are deleted. As shown, the GUI **400** can show a similar overview of the projects as the GUI **300**, described in conjunction with FIG. **3**, for example. At this point, the listing of projects in the deleted project page can be recovered in response to a recovery request. The GUI **400** can include an interactive element associated with each project to permanently remove or delete the respective project, such as deleting the project from the data repository **120**. The permanently deleted project may not be recoverable.

(63) Referring to FIG. **5**, the GUI **500** can represent a specification page to create a specification for the lighting system under a certain project. The GUI **500** can include various attributes selectable by the user via the GUI **500**. The attributes can be associated with interactive elements, such that the respective attributes can be selected in response to interaction with the interactive element (e.g., the data processing system **104** receiving an indication of interaction at a particular coordinate of the GUI **500**). The attributes can include at least luminescent types, mounting types, or luminescent features (e.g., sometimes referred to as features).

(64) For example, the luminescent types can include products **502** offered by the entity and the technology **504** related to the light fixtures. The product **502** can include at least one of compose light fixtures, graphite light fixtures, runway light fixtures, cove symmetric light fixtures, or other lighting designs. The product(s) **502** can be added, removed, or otherwise managed by the entity or administrator. Each type of the product **502** can be different from each other, although multiple types of products **502** can be used in the same project. The available products **502** presented to the user may be different depending on other selected attributes, such as selected luminescent features, mounting type, or technology **504**, for example.

(65) The technology **504** can include at least one of tunable lighting, static lighting, dim-warm lighting, sunset-dim lighting, or dim-candescent lighting, among other types of lighting. The technology **504** presented to the user may be managed by the entity. The types of technology **504** available can depend on the selected product **502**. The data processing system **104** (e.g., user interface generator **112**) can gray out, hide, or not display attributes (e.g., product **502**, technology **504**, etc.) that are unavailable for selection. The selection of the technology **504** can affect the availability of certain luminescent features, mounting types, or product **502**, for example.

(66) The GUI **500** can include an image **506** (e.g., overview) of the selected mounting type (if selected). Otherwise, the image **506** may include an example mounting type, a message indicating to select a mounting

type, or an empty space, among others depending on the configuration of the GUI **500**. The selection of the mounting type can be performed via interaction with the GUI **600**, as described in conjunction with FIG. **6**. The image **506** can be an interactive element configured to redirect the application to display GUI **600** for selecting or changing the mounting type, for example. The mounting type can include at least one of gypsum flat front (GFF), GFF wipe down, gypsum knife edge at a certain angle, wall mount, wall wash, axiom grazer, universal mount, or other types of mount for mounting the light fixtures. The selection of mounting types can be configurable or indicated by the entity. The GUI **600** can include selectable accessories **508** options, such as baffles or dust guards configured for the lighting system. The selection of the mounting type (or the luminescent type) can affect the availability of certain luminescent features.

(67) The GUI **500** can include selectable luminescent features in the luminaire specification section **510**. The luminescent features can include at least one of power consumption **512**, voltage **514**, CCT **516**, CRI **518**, beam **520**, finish **522**, control **524**, luminaire output **526**, or other features configurable for the lighting system. The power consumption **512** can represent an adjustable unit of energy per distance ratio (e.g., watts per foot) for the lighting system. In this case, the voltage **514** can represent the power output for powering the lighting system, which can be used for calculating the maximum run size given the power consumption **512**. The CCT **516** can represent the color temperature in kelvin (K), such as 2700K, 3000K, 3500K, 4000K, 5000K, etc. The CRI **518** can represent a quantitative measurement for rendering natural colors under an artificial white light source compared to sunlight from 0 (e.g., colors look the same) to 100 (e.g., reflects true color of the object). For example, the CRI **518** for the light fixtures can include at least one of a measurement of 82, 92, 98, etc. The beam **520** can represent the size of the light fixtures. The finish **522** can represent the color for the light fixtures. The control **524** can represent the type of software or hardware used to control the operation of the light fixtures (e.g., on/off, dim, color switching, etc.).

(68) The luminaire output **526** can include an interactive element for the user to enter a desired lumen per distance (e.g., foot) for the light fixtures. Depending on the lumen provided by the user, the data processing system **104** (e.g., light fixture selector **116**) can compute the corresponding power consumption **512** based on the selected CCT **516** and CRI **518**. For example, for purposes of the computation, individual CCTs can correspond to predefined multipliers. The higher the multiplier, the higher the power consumption **512**. An example of the multiplier corresponding to some CCTs can be shown in Table 1.

(69) TABLE-US-00001 TABLE 1 LUMEN FACTOR Output Factors CCT Multiplier 1850K 0.70 2000K 0.70 2200K 0.81 2500K 0.86 2700K 0.96 3000K 1.00 3500K 1.05 4000K 1.05 4500K 1.05 5000K 1.05 5700K 1.05 6500K 1.05

(70) Further, based on the selected CRI in addition to the selected CCT, the data processing system **104** can perform a lookup in a predefined table (e.g., Table 2) to determine the equivalent power consumption **512** necessary to obtain the desired luminaire output **526**. Selecting a relatively higher CRI can incur higher power consumption, as shown in Table 2. The tables can be predefined by the entity. The data processing system **104** can update the table in response to receiving update data from the manufacturing device **108**, for example. Although Table 2 illustrates the lumen data for a 3000K CCT with 92 CRI and 98 CRI, additional metrics or tables can be predefined for other CCTs or CRIs, and used in a similar manner to determine the corresponding power consumption given the desired luminaire output.

(71) TABLE-US-00002 3000K LUMEN DATA System Delivered Lumens/Ft Watts/Ft 92+ CRI 98+ CRI 2 135 119 3 228 200 4 286 253 5 401 354 6 504 445 7 606 534 8 733 645 9 970 849 10 1078 945 11 1174 1030 12 1277 1125

(72) After determining the equivalent unit of energy per distance ratio (e.g., wattage per foot), the data processing system **104** (e.g., layout configurator **114**) can determine the maximum size, length, or distance of a run (e.g., the distance of the light fixtures connected to the power feed). The data processing system **104** can use a predefined metric or table (e.g., Table 3) to determine the maximum run length based on the ratio and the voltage **514**. A relatively higher ratio can correspond to a relatively shorter maximum run length. A relatively lower ratio can correspond to a relatively longer maximum run length. Further, a relatively higher voltage **514** can provide a relatively longer maximum run length, and vice versa. In some cases, the maximum run length provided by Table 3 may represent the recommended maximum run length, as certain users may not operate the light fixtures (e.g., dimmable light fixtures) at the maximum supported luminaire output, for example.

(73) TABLE-US-00003 TABLE 3 MAX RUN PER POWER FEED Watts Maximum Run Length (Ft) Power 120VAC 277VAC 2 144 144 3 120 124 4 108 120 5 92 108 6 80 100 7 72 96 8 64 88 9 56 84 10 52 80 11 48 78 12 44 76

(74) The availability of the luminaire features of section **510** can be affected by the selection of other attributes. For instance, the data processing system **104** (e.g., user interface generator **112**) can update the GUI **500** to

change the availability of some luminaire features in response to selection or changes to the product **502**, technology **504**, mounting type, or certain other luminaire features. After the update, the data processing system **104** can render the GUI **500** that displays luminaire features of the lighting system that are compatible with other features and attributes. Hence, the data processing system **104** can eliminate errors in users selecting unavailable or non-compatible attributes. Further, depending on the selected attributes, the data processing system **104** can determine the available lengths corresponding to the attributes of the lighting system. The available lengths can be predefined by the entity and stored in the data repository **120**, each combination of attributes may be associated with a set of available lengths.

(75) The data processing system **104** can aggregate the selected attributes and store the selected attributes in the data repository **120** as a part of a configuration for the lighting system of the project. The data processing system **104** can save any updates to the configuration in real-time, periodically (e.g., every 10 seconds, 30 seconds, etc.), or upon interaction with a save button. The data processing system **104** can share or copy the configuration from the specification to other parts of the project, such as the quote or the drawings. Sharing the configuration can be performed automatically or in response to receiving a request from the user to share the configuration with other parts of the project. In this case, a similar configuration for the light fixtures can be reflected in at least the quote page or the drawing page.

(76) Referring to FIG. **6**, the GUI **600** can represent the specification creation page of GUI **500** with an expanded view for mounting type selection. The GUI **600** can include various elements or attributes presented in GUI **500**, such as the luminescent type, luminescent features, etc. The GUI **600** can include section **602** for selecting a mounting type. The section **602** can include images, texts, identification numbers, or other depictions of the various mounting types available for manufacturing or distribution by the entity. The GUI **600** can show available mounting types based on the selection of one or more other attributes, such as the luminescent type or luminescent features. Each depiction of the mounting type can correspond to an interactive element, where interaction with the interactive element can trigger a pop-up with additional information regarding the mounting type (e.g., hovering over the mounting type) or indicate a selection of the mounting type (e.g., mouse click or gesture on a touchscreen, etc.).

(77) Referring to FIG. **7**, the GUI **700** can represent the drawing page with a digital blueprint for configuring a layout for the installation of the lighting system. In some cases, prior to presenting GUI **700**, the data processing system **104** can prompt the user regarding whether to use at least one of the existing specifications (e.g., selected attributes) for the lighting system or select new attributes for the lighting system. In the case of using existing/saved attributes, the data processing system **104** can convert the selected attributes into the drawing in the digital blueprint.

(78) The GUI **700** can include selectable geometry **702** for the lighting system. Selecting the geometry **702** can allow the data processing system **104** (e.g., layout configurator **114**) to determine the style or design of the layout to generate in the drawing section **704**. The geometry **702** can include at least a straight run design, L-shape design, U-shape design, rectangular design, or custom design.

(79) The GUI **700** can include a drawing section **704** to provide the digital blueprint to the user. The drawing section **704** can illustratively present a physical space in which the lighting system is to be installed. When the user first navigates to the drawing page, the drawing section **704** can be empty or clear of any drawings (e.g., may include grids as a drawing guideline). As shown in GUI **700**, the data processing system **104** can receive a selected geometry **702** and coordinates input by the user via the GUI **700**. In response to detecting the coordinates, the data processing system **104** (e.g., layout configurator **114**) can generate the example layout in the drawing section **704**. The user can add new coordinates via input to the GUI **700** or change the relative location of the coordinates.

(80) For a predefined geometry **702**, the total coordinates may be predefined by the administrator, such as two coordinates for the straight run design, three coordinates for the L-shape design, four coordinates for the U-shape and rectangular designs, etc. In some cases, if one of the predefined geometry **702** is selected, the data processing system **104** (e.g., layout configurator **114**) can populate the drawing section **704** (e.g., digital blueprint) with an example layout. In certain cases, if the predefined geometry **702** is selected, the shape of the layout may be fixed, such that changing the coordinates can change at least one of the length, width, height, orientation, or positioning of the layout, without affecting the type of geometry **702**. In some other cases, adjusting the dimension (e.g., length, width, or height) or orientation of the layout can be performed using an object frame surrounding the layout or a rotational interactive element, for example. In this case, the data processing system **104** can change the selected geometry **702** to a custom geometry in response to adding or changing coordinate(s) in the digital blueprint.

(81) Although the example layout includes a single run of light fixtures, the data processing system **104** can

generate multiple runs of light fixtures based on the input coordinates and whether certain coordinates should be connected to other coordinates. Each of the runs can include light fixtures electrically coupled with each other. The light fixtures of different runs may or may not be electrically coupled with each other.

(82) The example layout in drawing section **704** can be configured as a U-shape design. The data processing system **104** (e.g., light fixture selector **116**) can use the model to select light fixtures having the selected attributes and conform with various constraints (e.g., gap constraint, power constraint, etc.). The attributes may be a part of the constraints. The GUI **700** can include a measurement of the lighting system for installation in the physical space. The lighting direction can be defaulted to illuminate towards the center of the U-shape design. The lighting direction can be reversed in response to interaction with the “reverse light” interactive element of the geometry **702**. The GUI **700** can include a corner light fixture if the “lit corner” option is selected in the geometry **702** option. Depending on certain attributes (e.g., luminescent type, mounting type, or luminescent features), the corner light fixture may not be available, hence the “lit corner” option may not be presented.

(83) The drawing section **704** can include selected light fixtures (e.g., selected by the data processing system **104**) positioned to conform with the layout. The data processing system **104** can use the model to select various straight lights (e.g., shown in the legend as “Straight Light” **1-4**) and L-shape lights (e.g., shown in the legend as “L-Shape Light”) positioned within a boundary. Each light fixture can include two ends. A first end can include a receiving end for receiving electricity, and a second end can include a plug end to connect one unit to another. At the end of a run, the data processing system **104** can use a light fixture including the receiving end and a voided end. The voided end may not include a plug to connect with another unit. The voided end can be configured to overlap a power feed, such that the gap constraint is satisfied when providing another run of light fixtures continuous from the prior run. The data processing system **104** can responsively analyze the constraints as the layout is adjusted, such that locations, lengths, or angles of the light fixtures can be modified or re-selected according to the adjusted layout while satisfying the constraints. Straight lights **1-4** may include different lengths or types of ends. For instance, the data processing system **104** can select the L-shape light for positioning at the corners of the U-shape design, the straight lights **1**, **3**, and **4** to satisfy the constraints and conform with the layout, and the straight light **2** (e.g., includes a voided end) for positioning at the end of the run. Additional details for each illustrated light fixture can be shown via interaction with the GUI **700**.

(84) Each light fixture can include a port for electrical connection with a battery for emergency power backup. When connected, the light fixture can be an emergency light. The data processing system **104** can receive input from the user of the client device **106** to insert or convert the existing light fixture to an emergency light. The data processing system **104** can receive input from the user to force power input (e.g., add power feed or move an existing power feed to the desired position). In response to inserting the power feed at the desired position according to the input (e.g., coordinate or position of the digital blueprint), the data processing system **104** can re-select the light fixtures. For example, prior to the power feed, the data processing system **104** can select a light fixture with a voided end. In another example, if the existing power feed is moved to another position between two light fixtures, the data processing system **104** can select two light fixtures with the voided end and position these light fixtures at the opposite ends of the run, in this case. The data processing system **104** can select the light fixtures that are available to avoid errors in the user selecting unavailable light fixtures.

(85) Referring to FIG. **8**, the GUI **800** can represent the drawing page with an overview of the light fixtures selected for the layout. In some cases, the GUI **800** can represent a quote page indicating materials (e.g., light fixtures) available for purchase to complete the project or lighting system. The GUI **800** can include a geometry run section **802** and a material section **804**. In this case, the geometry run section **802** can show the type of geometry **702** selected by the user (e.g., custom geometry), various light fixtures included selected by the data processing system **104** to conform with the layout, and an interactive element to redirect to the drawing section **704** (e.g., display GUI **700**). The material section **804** can include information related to the selected light fixtures of the layout, such as at least one of line number (e.g., label in the drawing for the respective part), the part number (e.g., serial number or identification of the light fixture), description, reference price, unit price, quantity, or total price, etc. The material section **804** can represent a portion of the quote for the user to order the light fixtures. The data processing system **104** can link the selected light fixtures to the quote page, thereby expediting the process to request the light fixtures from the entity.

(86) Referring to FIG. **9**, the GUI **900** can represent various projects (e.g., orders or sets of purchased light fixtures) historically purchased via the account used to access the application. The GUI **900** can include at least one of an overview of project information, dates (e.g., approval date, validation date, or shipment date, etc.), status of the order (e.g., starting, production, shipped, delivered, etc.), an indication of whether the order has been completed, or an interactive element to achieve the order, among others.

(87) Referring to FIG. 10, an example flow diagram of a method **1000** for manufacturing light fixtures. The example method **1000** can be executed, performed, or otherwise carried out by one or more components of the system **100** (e.g., data processing system **104**, client device **106**, manufacturing device **108**, etc.). The method **1000** can include identifying attributes, at ACT **1002**. At ACT **1004**, the method **1000** can include providing a digital blueprint. At ACT **1006**, the method **1000** can include detecting a layout. At ACT **1008**, the method **1000** can include selecting light fixtures. At ACT **1010**, the method **1000** can include determining whether the light fixtures exceed a power threshold. At ACT **1012**, the method **1000** can include adding a power feed. At ACT **1014**, the method **1000** can include rendering the light fixtures. At ACT **1016**, the method **1000** can include providing instruction for manufacturing light fixtures.

(88) Still referring to FIG. 10 in further detail, at ACT **1002**, the data processing system can identify attributes of a lighting system. The attributes can include at least one of a luminescent type, a mounting type, or a luminescent feature. The attributes can be adjusted by an administrator of the data processing system.

(89) At ACT **1004**, the data processing system can provide a digital blueprint of a physical space in which the lighting system is to be installed via a GUI. The data processing system can provide the digital blueprint via an application or a website accessible to a client device. The data processing system can manage the application. In some cases, the data processing system can provide the digital blueprint via a session hosted by the data processing system and established with the client device.

(90) The data processing system can detect, in response to input by a user using the client device via the GUI, various coordinates on the digital blueprint for installation of the lighting system. Based on the coordinates on the digital blueprint, the data processing system can generate a layout including one or more interconnecting lines or conjunction between one or more combinations of the coordinates. The data processing system can detect changes to the layout in response to input via the GUI, such as adding, moving, or removing coordinates from the digital blueprint. In some cases, the changes to the layout can include changing the lighting direction, managing power feed, managing emergency light fixtures/units, resizing the shape of the layout, managing corner lighting, changing the selection of certain attributes, etc.

(91) At ACT **1006**, the data processing system can detect the layout for installation of the lighting system responsive to the input via the GUI. The data processing system can detect the various portions of the layout as the coordinates are provided in the digital blueprint. The data processing system can detect a new layout in response to the input or an existing layout retrieved from a data repository via the input, for example.

(92) At ACT **1008**, the data processing system can select one or more light fixtures using a model. The data processing system can use the model configured to satisfy a gap constraint, among other constraints. The data processing system can use the model to select the light fixtures based on the layout and the attributes selected by the user, such that the gap constraint is satisfied when the light fixtures are electrically coupled to conform to the layout. The gap constraint can be predefined by the administrator, such as 0.5 inches, 0.4 inches, 0.3 inches, etc. The data processing system can select the size (e.g., length) of the light fixtures having the attributes and position of the selected light fixtures to conform with the layout, such that the gap between the light fixtures are at or below the gap constraint (e.g., maximum gap threshold).

(93) In some cases, the data processing system can select the model from various models stored in the data repository based on at least one of the attributes selected by the user. In this case, the data processing system can use the model to select light fixtures that have the attribute(s) selected by the user. In certain cases, the data processing system can select the model from various models based on at least one or a combination of the luminescent type, mounting type, or luminescent feature. In this case, the data processing system can use the model to select the light fixtures having one or a combination of the various attributes (e.g., luminescent type, mounting type, or luminescent feature).

(94) The data processing system can select the light fixtures including a first light fixture and a second light fixture. The first and second light fixtures can have different configurations from each other. The configuration can include at least one of the length or the shape of the light fixtures. The availability of the length (or shape) of the light fixtures can depend on at least one of the attributes selected. The length may be predefined by an entity manufacturing or distributing the light fixtures, such as 4 inches, 8 inches, 12 inches, 16 inches, 18 inches, 32 inches, 48 inches, etc. The shape may include at least a straight light unit, an L-shape light unit, or other angles of light units.

(95) In some cases, the data processing system can detect, responsive to input via the GUI, the layout (e.g., coordinates entered via the GUI) including a first dimension and a second dimension. The second dimension can be perpendicular with the first dimension. In this case, the data processing system can select light fixtures including a first, a second, and a third light fixture. The first light fixture can have a first run size and be a straight run (e.g., straight shape/design). The second light fixture can have a second run size and be an L-shape.

The third light fixture can have a third size and be oriented straight run. To conform with the layout, the data processing system can (e.g., digitally) electrically couple the first light fixture to the third light fixture via the second light fixture in the digital blueprint, such that the first light fixture is perpendicular to the second light fixture.

(96) In some cases, the data processing system can identify attributes including an emergency lighting type. The data processing system can detect, responsive to input via the GUI, an indication of emergency lighting type selection and associated coordinates. The data processing system can select an emergency light unit for the digital blueprint to conform with the layout. In some cases, if the coordinate(s) correspond to an existing light fixture, the data processing system can select the existing light fixture and update the selected light fixture to the emergency light unit for connection with a battery source.

(97) The data processing system can identify the direction for lighting and a number of power feeds for the layout. The data processing system can select the light fixtures based on the direction and the number of power feeds, such as the selection of light fixtures with a voided end or light fixtures for lighting in a certain direction. The data processing system can identify the direction of lighting based on selected attribute(s) or a user-selected geometry for the lighting system, for example.

(98) In some other cases, the data processing system can select the model to satisfy a unit of energy per distance ratio. In this case, the data processing system can use the model to determine whether the run of light fixtures satisfies the unit of energy per distance ratio (e.g., power constraint, such as watts per foot of the light fixture).

(99) At ACT **1010**, the data processing system can determine whether the light fixtures, selected via the model, exceed a power threshold (e.g., power constraint). The data processing system can proceed to ACT **1012** if the light fixtures exceed the power threshold. The data processing system can proceed to ACT **1014** if the light fixtures do not exceed the power threshold. The power threshold can be based on the unit of energy per distance ratio according to the selected attributes, such as at least one of the desired luminaire output, CCT, CRI, outlet voltage, etc. For instance, the selected attributes can affect the wattage per foot for operating the light fixtures. The wattage per foot can indicate the maximum distance of light fixtures that can be powered given the outlet voltage (e.g., maximum power feed output).

(100) At ACT **1012**, the data processing system can add a second power feed responsive to determining that the light fixtures exceed the power threshold. The data processing system can add the second power feed at the end of a first set of light fixtures (e.g., first run) that exceeded the power threshold, such that a second set of light fixtures (e.g., second run) can be positioned continuing from the first set of light fixtures and powered with the second power feed.

(101) At ACT **1014**, the data processing system can select at least a portion of the light fixtures responsive to detecting, via the GUI, at least a portion of the layout for installation prior to completion of the layout for installation. The data processing system can provide the portion of the light fixtures, prior to the completion of the layout for installation, for rendering via the GUI. The data processing system can continue to provide various portions of the light fixtures for rendering via the GUI before and after completion. The data processing system can provide an updated portion(s) of the layout for rendering via the GUI.

(102) At ACT **1016**, the data processing system can provide instruction to a manufacturing device to cause the manufacturing of the selected light fixtures in accordance with the layout and the plurality of attributes. In some cases, the data processing system can provide the instruction to cause a distribution of the selected light fixtures, if the light fixtures are available in the inventory of the manufacturer or distributor, for example.

Accordingly, the data processing system can enable the user to visualize the layout for the installation of the lighting system and select desired attributes of the lighting system without compatibility or availability errors.

(103) FIG. **11** is a block diagram of an example computer system **1100**. The computer system or computing device **1100** can include or be used to implement the data processing system **104**, or its components such as the data processing system **104**. The computing system **1100** includes at least one bus **1105** or other communication component for communicating information and at least one processor **1110** or processing circuit coupled to the bus **1105** for processing information. The computing system **1100** can also include one or more processors **1110** or processing circuits coupled to the bus for processing information. The computing system **1100** also includes at least one main memory **1115**, such as a random access memory (RAM) or other dynamic storage device, coupled to the bus **1105** for storing information, and instructions to be executed by the processor **1110**. The main memory **1115** can also be used for storing position information, utility grid data, command instructions, device status information, environmental information within or external to the utility grid, information on characteristics of electricity, or other information during execution of instructions by the processor **1110**. The computing system **1100** may further include at least one read only memory (ROM) **1120** or

other static storage device coupled to the bus **1105** for storing static information and instructions for the processor **1110**. A storage device **1125**, such as a solid state device, magnetic disk or optical disk, can be coupled to the bus **1105** to persistently store information and instructions.

(104) The computing system **1100** may be coupled via the bus **1105** to a display **1135**, such as a liquid crystal display, or active matrix display, for displaying information to a user such as an administrator of the data processing system or the utility grid. An input device **1130**, such as a keyboard or voice interface may be coupled to the bus **1105** for communicating information and commands to the processor **1110**. The input device **1130** can include a touch screen display **1135**. The input device **1130** can also include a cursor control, such as a mouse, a trackball, or cursor direction keys, for communicating direction information and command selections to the processor **1110** and for controlling cursor movement on the display **1135**. The display **1135** can be part of the data processing system **104**, or other component of FIG. 1.

(105) The processes, systems and methods described herein can be implemented by the computing system **1100** in response to the processor **1110** executing an arrangement of instructions contained in main memory **1115**. Such instructions can be read into main memory **1115** from another computer-readable medium, such as the storage device **1125**. Execution of the arrangement of instructions contained in main memory **1115** causes the computing system **1100** to perform the illustrative processes described herein. One or more processors in a multi-processing arrangement may also be employed to execute the instructions contained in main memory **1115**. Hard-wired circuitry can be used in place of or in combination with software instructions together with the systems and methods described herein. Systems and methods described herein are not limited to any specific combination of hardware circuitry and software.

FURTHER EXAMPLES

(106) The following examples pertain to further embodiments, from which numerous permutations and configurations will be apparent.

(107) Example 1 includes a method of manufacturing light fixtures, comprising: identifying, by a data processing system comprising one or more processors and memory, a plurality of attributes of a lighting system, the plurality of attributes comprising at least one of a luminescent type, a mounting type, or a luminescent feature; detecting, by the data processing system responsive to input via a graphical user interface, a layout for installation of the lighting system; selecting, by the data processing system using a model configured to satisfy a gap constraint, a plurality of light fixtures based on the layout and the plurality of attributes that, when electrically coupled to conform to the layout, satisfy the gap constraint; and providing, by the data processing system, an instruction to cause manufacturing of the selected plurality of light fixtures in accordance with the layout and the plurality of attributes.

(108) Example 2 includes the subject matter of Example 1, comprising: providing, by the data processing system via the graphical user interface, a digital blueprint of a physical space in which the lighting system is to be installed; detecting, by the data processing system responsive to input via the graphical user interface, a plurality of coordinates on the digital blueprint for installation of the lighting system; and generating, by the data processing system, the layout based on the plurality of coordinates on the digital blueprint.

(109) Example 3 includes the subject matter of any of Examples 1 and 2, comprising: detecting, by the data processing system responsive to input via the graphical user interface, the layout comprising a first dimension and a second dimension that is perpendicular with the first dimension; and selecting, by the data processing system, the plurality of light fixtures comprising a first light fixture with a first run size and a straight run, a second light fixture with a second run size and an L-shape, and a third light fixture with a third size and the straight run, wherein the first light fixture is electrically coupled to the third light fixture via the second light fixture.

(110) Example 4 includes the subject matter of any of Examples 1 through 3, comprising: selecting, by the data processing system, the model from a plurality of models based on at least one of the plurality of attributes.

(111) Example 5 includes the subject matter of any of Examples 1 through 4, comprising: selecting, by the data processing system, the model from a plurality of models based on the mounting type and the luminescent feature.

(112) Example 6 includes the subject matter of any of Examples 1 through 5, comprising: selecting, by the data processing system, the model to satisfy a unit of energy per distance ratio.

(113) Example 7 includes the subject matter of any of Examples 1 through 6, comprising: selecting, by the data processing system, the plurality of light fixtures comprising a first light fixture and a second light fixture having a different configuration than the first light fixture.

(114) Example 8 includes the subject matter of any of Examples 1 through 7, comprising: wherein the different configuration comprises at least one of a length or a shape.

(115) Example 9 includes the subject matter of any of Examples 1 through 8, comprising: selecting, by the data processing system, at least a portion of the plurality of light fixtures responsive to detecting, via the graphical user interface, at least a portion of the layout for installation prior to completion of the layout for installation; and providing, by the data processing system, for render via the graphical user interface prior to completion of the layout for installation, the at least the portion of the plurality of light fixtures.

(116) Example 10 includes the subject matter of any of Examples 1 through 9, comprising: identifying, by the data processing system, the plurality of attributes comprising an emergency lighting type.

(117) Example 11 includes the subject matter of any of Examples 1 through 10, comprising: identifying, by the data processing system, a direction for lighting and a number of power feeds for the layout; and selecting, by the data processing system, the plurality of light fixtures based on the direction and the number of power feeds.

(118) Example 12 includes the subject matter of any of Examples 1 through 11, comprising: determining, by the data processing system, the plurality of light fixtures selected via the model exceeds a power threshold; and adding, by the data processing system, a second power feed responsive to the determination.

(119) Example 13 includes a system comprising: a data processing system comprising one or more processors, coupled with memory, to: identify a plurality of attributes of a lighting system, the plurality of attributes comprising at least one of a luminescent type, a mounting type, or a luminescent feature; detect, responsive to input via a graphical user interface, a layout for installation of the lighting system; select, via a model configured to satisfy a gap constraint, a plurality of light fixtures based on the layout and the plurality of attributes that, when electrically coupled to conform to the layout, satisfy the gap constraint; and provide an instruction to cause manufacturing of the selected plurality of light fixtures in accordance with the layout and the plurality of attributes.

(120) Example 14 includes the subject matter of Example 13, wherein the data processing system is further configured to: provide, via the graphical user interface, a digital blueprint of a physical space in which the lighting system is to be installed; detect, responsive to input via the graphical user interface, a plurality of coordinates on the digital blueprint for installation of the lighting system; and generate the layout based on the plurality of coordinates on the digital blueprint.

(121) Example 15 includes the subject matter of any of Examples 13 and 14, wherein the data processing system is further configured to: detect, responsive to input via the graphical user interface, the layout comprising a first dimension and a second dimension that is perpendicular with the first dimension; and select the plurality of light fixtures comprising a first light fixture with a first run size and a straight run, a second light fixture with a second run size and an L-shape, and a third light fixture with a third size and the straight run, wherein the first light fixture is electrically coupled to the third light fixture via the second light fixture.

(122) Example 16 includes the subject matter of any of Examples 13 through 15, wherein the data processing system is further configured to: select the model from a plurality of models based on at least one of the plurality of attributes.

(123) Example 17 includes the subject matter of any of Examples 13 through 16, wherein the data processing system is further configured to: select the plurality of light fixtures comprising a first light fixture and a second light fixture having a different configuration than the first light fixture.

(124) Example 18 includes the subject matter of any of Examples 13 through 17, wherein the data processing system is further configured to: determine the plurality of light fixtures selected via the model exceeds a power threshold; and add a second power feed responsive to the determination.

(125) Example 19 includes a non-transitory computer readable storage medium storing instructions that, when executed by one or more processors, cause the one or more processors to: identify a plurality of attributes of a lighting system, the plurality of attributes comprising at least one of a luminescent type, a mounting type, or a luminescent feature; detect, responsive to input via a graphical user interface, a layout for installation of the lighting system; select, via a model configured to satisfy a gap constraint, a plurality of light fixtures based on the layout and the plurality of attributes that, when electrically coupled to conform to the layout, satisfy the gap constraint; and provide an instruction to cause manufacturing of the selected plurality of light fixtures in accordance with the layout and the plurality of attributes.

(126) Example 20 includes the subject matter of Example 19, wherein the instructions further comprise instructions to: provide, via the graphical user interface, a digital blueprint of a physical space in which the lighting system is to be installed; detect, responsive to input via the graphical user interface, a plurality of coordinates on the digital blueprint for installation of the lighting system; and generate the layout based on the plurality of coordinates on the digital blueprint.

(127) Although an example computing system has been described in FIG. 7, the subject matter including the operations described in this specification can be implemented in other types of digital electronic circuitry, or in

computer software, firmware, or hardware, including the structures disclosed in this specification and their structural equivalents, or in combinations of one or more of them.

(128) Some of the descriptions herein emphasize the structural independence of the aspects of the system components (e.g., arbitration component) and illustrate one grouping of operations and responsibilities of these system components. Other groupings that execute similar overall operations are understood to be within the scope of the present application. Modules can be implemented in hardware or as computer instructions on a non-transient computer-readable storage medium, and modules can be distributed across various hardware- or computer-based components.

(129) The systems described above can provide multiple ones of any or each of those components and these components can be provided on either a standalone system or on multiple instantiation in a distributed system. In addition, the systems and methods described above can be provided as one or more computer-readable programs or executable instructions embodied on or in one or more articles of manufacture. The article of manufacture can be cloud storage, a hard disk, a CD-ROM, a flash memory card, a PROM, a RAM, a ROM, or a magnetic tape. In general, the computer-readable programs can be implemented in any programming language, such as LISP, PERL, C, C++, C #, PROLOG, or in any byte code language such as JAVA. The software programs or executable instructions can be stored on or in one or more articles of manufacture as object code.

(130) Example and non-limiting module implementation elements include sensors providing any value determined herein, sensors providing any value that is a precursor to a value determined herein, datalink or network hardware including communication chips, oscillating crystals, communication links, cables, twisted pair wiring, coaxial wiring, shielded wiring, transmitters, receivers, or transceivers, logic circuits, hard-wired logic circuits, reconfigurable logic circuits in a particular non-transient state configured according to the module specification, any actuator including at least an electrical, hydraulic, or pneumatic actuator, a solenoid, an op-amp, analog control elements (springs, filters, integrators, adders, dividers, gain elements), or digital control elements.

(131) The subject matter and the operations described in this specification can be implemented in digital electronic circuitry, or in computer software, firmware, or hardware, including the structures disclosed in this specification and their structural equivalents, or in combinations of one or more of them. The subject matter described in this specification can be implemented as one or more computer programs, e.g., one or more circuits of computer program instructions, encoded on one or more computer storage media for execution by, or to control the operation of, data processing apparatuses. Alternatively or in addition, the program instructions can be encoded on an artificially generated propagated signal, e.g., a machine-generated electrical, optical, or electromagnetic signal that is generated to encode information for transmission to suitable receiver apparatus for execution by a data processing apparatus. A computer storage medium can be, or be included in, a computer-readable storage device, a computer-readable storage substrate, a random or serial access memory array or device, or a combination of one or more of them. While a computer storage medium is not a propagated signal, a computer storage medium can be a source or destination of computer program instructions encoded in an artificially generated propagated signal. The computer storage medium can also be, or be included in, one or more separate components or media (e.g., multiple CDs, disks, or other storage devices include cloud storage). The operations described in this specification can be implemented as operations performed by a data processing apparatus on data stored on one or more computer-readable storage devices or received from other sources.

(132) The terms “computing device”, “component” or “data processing apparatus” or the like encompass various apparatuses, devices, and machines for processing data, including by way of example a programmable processor, a computer, a system on a chip, or multiple ones, or combinations of the foregoing. The apparatus can include special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit). The apparatus can also include, in addition to hardware, code that creates an execution environment for the computer program in question, e.g., code that constitutes processor firmware, a protocol stack, a database management system, an operating system, a cross-platform runtime environment, a virtual machine, or a combination of one or more of them. The apparatus and execution environment can realize various different computing model infrastructures, such as web services, distributed computing and grid computing infrastructures.

(133) A computer program (also known as a program, software, software application, app, script, or code) can be written in any form of programming language, including compiled or interpreted languages, declarative or procedural languages, and can be deployed in any form, including as a stand-alone program or as a module, component, subroutine, object, or other unit suitable for use in a computing environment. A computer program

can correspond to a file in a file system. A computer program can be stored in a portion of a file that holds other programs or data (e.g., one or more scripts stored in a markup language document), in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, sub programs, or portions of code). A computer program can be deployed to be executed on one computer or on multiple computers that are located at one site or distributed across multiple sites and interconnected by a communication network.

(134) The processes and logic flows described in this specification can be performed by one or more programmable processors executing one or more computer programs to perform actions by operating on input data and generating output. The processes and logic flows can also be performed by, and apparatuses can also be implemented as, special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit). Devices suitable for storing computer program instructions and data can include non-volatile memory, media and memory devices, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto optical disks; and CD ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.

(135) The subject matter described herein can be implemented in a computing system that includes a back end component, e.g., as a data server, or that includes a middleware component, e.g., an application server, or that includes a front end component, e.g., a client computer having a graphical user interface or a web browser through which a user can interact with an implementation of the subject matter described in this specification, or a combination of one or more such back end, middleware, or front end components. The components of the system can be interconnected by any form or medium of digital data communication, e.g., a communication network. Examples of communication networks include a local area network (“LAN”) and a wide area network (“WAN”), an inter-network (e.g., the Internet), and peer-to-peer networks (e.g., ad hoc peer-to-peer networks).

(136) While operations are depicted in the drawings in a particular order, such operations are not required to be performed in the particular order shown or in sequential order, and all illustrated operations are not required to be performed. Actions described herein can be performed in a different order.

(137) Having now described some illustrative implementations, it is apparent that the foregoing is illustrative and not limiting, having been presented by way of example. In particular, although many of the examples presented herein involve specific combinations of method acts or system elements, those acts and those elements may be combined in other ways to accomplish the same objectives. Acts, elements and features discussed in connection with one implementation are not intended to be excluded from a similar role in other implementations or implementations.

(138) The phraseology and terminology used herein is for the purpose of description and should not be regarded as limiting. The use of “including” “comprising” “having” “containing” “involving” “characterized by” “characterized in that” and variations thereof herein, is meant to encompass the items listed thereafter, equivalents thereof, and additional items, as well as alternate implementations consisting of the items listed thereafter exclusively. In one implementation, the systems and methods described herein consist of one, each combination of more than one, or all of the described elements, acts, or components.

(139) Any references to implementations or elements or acts of the systems and methods herein referred to in the singular may also embrace implementations including a plurality of these elements, and any references in plural to any implementation or element or act herein may also embrace implementations including only a single element. References in the singular or plural form are not intended to limit the presently disclosed systems or methods, their components, acts, or elements to single or plural configurations. References to any act or element being based on any information, act or element may include implementations where the act or element is based at least in part on any information, act, or element.

(140) Any implementation disclosed herein may be combined with any other implementation or embodiment, and references to “an implementation,” “some implementations,” “one implementation” or the like are not necessarily mutually exclusive and are intended to indicate that a particular feature, structure, or characteristic described in connection with the implementation may be included in at least one implementation or embodiment. Such terms as used herein are not necessarily all referring to the same implementation. Any implementation may be combined with any other implementation, inclusively or exclusively, in any manner consistent with the aspects and implementations disclosed herein.

(141) References to “or” may be construed as inclusive so that any terms described using “or” may indicate any of a single, more than one, and all of the described terms. For example, a reference to “at least one of ‘A’ and ‘B’” can include only ‘A’, only ‘B’, as well as both ‘A’ and ‘B’. Such references used in conjunction with “comprising” or other open terminology can include additional items.

(142) Where technical features in the drawings, detailed description or any claim are followed by reference signs, the reference signs have been included to increase the intelligibility of the drawings, detailed description, and claims. Accordingly, neither the reference signs nor their absence have any limiting effect on the scope of any claim elements.

(143) Modifications of described elements and acts such as variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations can occur without materially departing from the teachings and advantages of the subject matter disclosed herein. For example, elements shown as integrally formed can be constructed of multiple parts or elements, the position of elements can be reversed or otherwise varied, and the nature or number of discrete elements or positions can be altered or varied. Other substitutions, modifications, changes and omissions can also be made in the design, operating conditions and arrangement of the disclosed elements and operations without departing from the scope of the present disclosure.

(144) The systems and methods described herein may be embodied in other specific forms without departing from the characteristics thereof. Scope of the systems and methods described herein is thus indicated by the appended claims, rather than the foregoing description, and changes that come within the meaning and range of equivalency of the claims are embraced therein.

(145) While this specification contains many specific implementation details, these should not be construed as limitations on the scope of any inventions or of what can be claimed, but rather as descriptions of features specific to particular embodiments of particular aspects. Certain features described in this specification in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable subcombination. Moreover, although features can be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination can be directed to a subcombination or variation of a subcombination.

(146) Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing can be advantageous. Moreover, the separation of various system components in the embodiments described above should not be understood as requiring such separation in all embodiments, and it should be understood that the described program components and systems can generally be integrated in a single software product or packaged into multiple software products.

(147) Thus, particular embodiments of the subject matter have been described. In some cases, the actions recited in the claims can be performed in a different order and still achieve desirable results. In addition, the processes depicted in the accompanying figures do not necessarily require the particular order shown, or sequential order, to achieve desirable results.

Claims

1. A method of manufacturing light fixtures, comprising: identifying, by a data processing system comprising one or more processors and memory, a plurality of attributes of a lighting system, the plurality of attributes comprising at least one of a luminescent type, a mounting type, or a luminescent feature; detecting, by the data processing system responsive to input via a graphical user interface, a layout for installation of the lighting system; selecting, by the data processing system using a model configured to satisfy a gap constraint, a plurality of light fixtures based on the layout and the plurality of attributes that, when electrically coupled to conform to the layout, satisfy the gap constraint; and providing, by the data processing system, an instruction to cause manufacturing of the selected plurality of light fixtures in accordance with the layout and the plurality of attributes.
2. The method of claim 1, comprising: providing, by the data processing system via the graphical user interface, a digital blueprint of a physical space in which the lighting system is to be installed; detecting, by the data processing system responsive to input via the graphical user interface, a plurality of coordinates on the digital blueprint for installation of the lighting system; and generating, by the data processing system, the layout based on the plurality of coordinates on the digital blueprint.
3. The method of claim 1, comprising: detecting, by the data processing system responsive to input via the graphical user interface, the layout comprising a first dimension and a second dimension that is perpendicular with the first dimension; and selecting, by the data processing system, the plurality of light fixtures comprising

a first light fixture with a first run size and a straight run, a second light fixture with a second run size and an L-shape, and a third light fixture with a third size and the straight run, wherein the first light fixture is electrically coupled to the third light fixture via the second light fixture.

4. The method of claim 1, comprising: selecting, by the data processing system, the model from a plurality of models based on at least one of the plurality of attributes.

5. The method of claim 1, comprising: selecting, by the data processing system, the model from a plurality of models based on the mounting type and the luminescent feature.

6. The method of claim 1, comprising: selecting, by the data processing system, the model to satisfy a unit of energy per distance ratio.

7. The method of claim 1, comprising: selecting, by the data processing system, the plurality of light fixtures comprising a first light fixture and a second light fixture having a different configuration than the first light fixture.

8. The method of claim 7, comprising: wherein the different configuration comprises at least one of a length or a shape.

9. The method of claim 1, comprising: selecting, by the data processing system, at least a portion of the plurality of light fixtures responsive to detecting, via the graphical user interface, at least a portion of the layout for installation prior to completion of the layout for installation; and providing, by the data processing system, for render via the graphical user interface prior to completion of the layout for installation, the at least the portion of the plurality of light fixtures.

10. The method of claim 1, comprising: identifying, by the data processing system, the plurality of attributes comprising an emergency lighting type.

11. The method of claim 1, comprising: identifying, by the data processing system, a direction for lighting and a number of power feeds for the layout; and selecting, by the data processing system, the plurality of light fixtures based on the direction and the number of power feeds.

12. The method of claim 1, comprising: determining, by the data processing system, the plurality of light fixtures selected via the model exceeds a power threshold; and adding, by the data processing system, a second power feed responsive to the determination.

13. A system, comprising: a data processing system comprising one or more processors, coupled with memory, to: identify a plurality of attributes of a lighting system, the plurality of attributes comprising at least one of a luminescent type, a mounting type, or a luminescent feature; detect, responsive to input via a graphical user interface, a layout for installation of the lighting system; select, via a model configured to satisfy a gap constraint, a plurality of light fixtures based on the layout and the plurality of attributes that, when electrically coupled to conform to the layout, satisfy the gap constraint; and provide an instruction to cause manufacturing of the selected plurality of light fixtures in accordance with the layout and the plurality of attributes.

14. The system of claim 13, wherein the data processing system is further configured to: provide, via the graphical user interface, a digital blueprint of a physical space in which the lighting system is to be installed; detect, responsive to input via the graphical user interface, a plurality of coordinates on the digital blueprint for installation of the lighting system; and generate the layout based on the plurality of coordinates on the digital blueprint.

15. The system of claim 13, wherein the data processing system is further configured to: detect, responsive to input via the graphical user interface, the layout comprising a first dimension and a second dimension that is perpendicular with the first dimension; and select the plurality of light fixtures comprising a first light fixture with a first run size and a straight run, a second light fixture with a second run size and an L-shape, and a third light fixture with a third size and the straight run, wherein the first light fixture is electrically coupled to the third light fixture via the second light fixture.

16. The system of claim 13, wherein the data processing system is further configured to: select the model from a plurality of models based on at least one of the plurality of attributes.

17. The system of claim 13, wherein the data processing system is further configured to: select the plurality of light fixtures comprising a first light fixture and a second light fixture having a different configuration than the first light fixture.

18. The system of claim 13, wherein the data processing system is further configured to: determine the plurality of light fixtures selected via the model exceeds a power threshold; and add a second power feed responsive to the determination.

19. A non-transitory computer readable storage medium storing instructions that, when executed by one or more processors, cause the one or more processors to: identify a plurality of attributes of a lighting system, the plurality of attributes comprising at least one of a luminescent type, a mounting type, or a luminescent feature;

detect, responsive to input via a graphical user interface, a layout for installation of the lighting system; select, via a model configured to satisfy a gap constraint, a plurality of light fixtures based on the layout and the plurality of attributes that, when electrically coupled to conform to the layout, satisfy the gap constraint; and provide an instruction to cause manufacturing of the selected plurality of light fixtures in accordance with the layout and the plurality of attributes.

20. The non-transitory computer readable storage medium of claim 19, wherein the instructions further comprise instructions to: provide, via the graphical user interface, a digital blueprint of a physical space in which the lighting system is to be installed; detect, responsive to input via the graphical user interface, a plurality of coordinates on the digital blueprint for installation of the lighting system; and generate the layout based on the plurality of coordinates on the digital blueprint.
